Week 8

Safety, and the Halfway Mark

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Today

0:00 – 1:15	★ **Presentations** — 12 min + 5 min questions, per team
1:15 – 1:25	Break
1:25 – 2:20	Lecture — safety, authority, failsafe
2:20 – 2:45	Cross-team dependency review
2:45 – 3:00	Peer evaluation 1

Presentations first, while everyone is fresh.

The safety lecture lands afterwards **on purpose.**

Presentation rubric — 100 points

	Points	
Subsystem demonstrated	30	Live or recorded. **Partial is expected**
Evidence quality	25	Numbers, plots, logs. Measured, not asserted
Interface clarity	20	What you need, from whom, **by when**
Risk honesty	15	What is going wrong
Delivery	10	Clear, on time, everyone speaks

Present to your classmates, not to me. They are the ones who have to merge with you.

The two ways teams lose points here

Presenting a plan instead of a result.

You are three weeks in. **"We researched the options"** is a Week 6 answer.
Show what you built — even small, even half-broken.

Not having talked to the track you depend on.

Interface clarity is 20 points precisely because it fails silently until Week 13.
If your slide says **"we need the measured wheelbase"** and Chassis has never heard that, **both** teams lose points.

A team that says "our angle sensor reads garbage past 300° and we do not yet know why" scores above one with a clean demo it cannot explain.

Risk honesty is 15 points, and it is free. Nobody expects a working subsystem at Week 8.

Safety

Authority is a ladder, and it is electrical

Layer	Works if the firmware has crashed?
E-stop — hardwired contactor cuts traction	**Yes**
Relay MUX — coil drops → stock steering	**Yes**
RC signal MUX — hardware, on the servo pulse	**Yes**
SBUS override — into the Teensy	No

Default is STOCK

On **every** power-up. On **every** fault.

The MUX holds the DBW position only while energised,
so **losing power is reverting to manual.**

Safety is the **resting state** — not a state you have to reach.

The failsafe matrix

Loss	Detection	Behaviour
Command loss	setpoint stale > 500 ms	ESTOP, throttle → 0, steering centred then de-energised
USB link loss	no micro-ROS session	Teensy enters ESTOP autonomously
RC loss	SBUS frame timeout	ESTOP
Brownout	logic-rail undervoltage	MUX coil drops → **STOCK**
E-stop pressed	hardwired	traction cut, MUX → STOCK
Sabertooth command loss	serial timeout	motors stop
Steering at limit	stall detected	clamp effort **toward centre only**
Firmware hang	serial timeout + watchdog	motors stop, Teensy reset to neutral
Angle sensor fault	plausibility check each loop	ESTOP — **never run the loop on a bad angle**

One pattern: **detect within a bounded time, resolve to a defined safe state, require explicit re-arm.**

Row 2 is a regression, and the record says so

Under the **old** Pixhawk design, USB carried **feedback only** —
the setpoint arrived separately as servo PWM.
A USB dropout left steering **still tracking**.

Under **D3**, the same link carries the setpoint.
A USB dropout **removes** it.

That is worse in one specific respect, and it was accepted **knowingly.**

Why it was accepted

The failure is now detected **in one place with one timeout**, and resolves to a **defined safe state** — rather than to

> **"keep tracking a setpoint whose author is gone"**

A stale setpoint with a live actuator is more dangerous than a stop.

And it comes with a measurement: log USB stability over ≥ 30 min.

If dropouts occur at all, treat it as a blocking defect, not a nuisance.

The state machine

```
┌─────────┐    fault / stale / RC loss     ┌──────────────┐
│  ESTOP  │◄──────────────────────────────│  AUTONOMOUS  │
└────┬────┘                                └──────▲───────┘
     │ explicit re-arm                            │ operator hands over
     ▼                                            │
┌──────────────┐                                  │
│  MANUAL_RC   │ ─────────────────────────────────┘
└──────────────┘
```

- **Entered automatically. Left only explicitly.** No fault clears itself into motion
- **Any set fault bit is a reason to refuse to leave ESTOP** — which is why the bitfield names six causes rather than being a boolean

FMEA: the one that scores severity 5

> ***Magnet wrap*** *— the sensed shaft rotates past one turn.*
> *The angle folds back, and the loop drives the wrong way,* ***silently****.*

Note **why** it is a 5. Not because the consequence is worse than the others.

Because it is **silent**. The sensor reads a perfectly plausible value.
Nothing detects it. The loop does exactly what it was told —
in the wrong direction.

The mitigation is mechanical

> ***Load-side mounting (±22.5°) makes wrap mechanically impossible.***

A software check would have to distinguish a wrapped reading from a legitimate one — and for a rotary sensor those can be **identical**.

So the design **eliminates the failure mode instead of detecting it.**

When you cannot reliably detect a failure, look for a way to make it unreachable.

You have now seen this move twice: the Lisbon declination in Week 6, and magnet wrap here.

Staged bring-up

> ***Bench before vehicle; wheels-off before wheels-on; walking pace before anything faster.***
> *No stage begins until the previous stage passes.*

Stage	What	Gate
0	Bench, no motors	≥ 50 Hz status; **zero USB dropouts over 30 min**
1	Bench, steering motor only	≥ 200 Hz loop, no runaway, freewheel on power cut. **E4 decision point**
2	Bench, both channels + RC	**Both override layers — incl. the MUX with the Teensy deliberately halted**
3	Relay MUX + E-stop, wheels off	Every failsafe row, incl. E-stop **with the laptop powered off**
4	Vehicle, wheels on stands	Full integration; no brownout under steering stall
5	Ground, walking pace, operator alongside	E-stop on the move. **Only then** does autonomy engage

This is a hard gate in this course

A team that skips a stage does not lose marks.

It does not demo.

The protocol is staged specifically so a class can work on a vehicle heavy enough to hurt someone **without an instructor watching every action.**

That only works if nobody treats it as optional.

Stage 2 in particular: the hardware MUX must be shown **with the Teensy halted.** The record is explicit — **demonstrated, not assumed.**

Cross-team dependency review

Every ask, from every charter, on one board.

- Who needs what, from whom, by when
- **Is the date still real?**
- **What is nobody claiming?** Gaps between charters are how Week 13 fails

If your track will miss a merge date, **say so now.**

Announcing a slip at Week 8 is professional. Discovering it at Merge 3 is not.

Peer evaluation 1

Confidential. 1–5 on contribution, reliability, collaboration — including yourself.

Plus free text: **what did this person do that the git history would not show me?**

Documentation, debugging someone else's problem, and asking the question that saved a week are all real work that leaves no commit.

Can adjust an individual's team-component grade by **±15%**.

If your team has a problem

Now is when it is fixable.

At Week 8 there are seven weeks left.

At Week 15 there is nothing I can do but adjust grades —
which helps nobody build anything.

Next up: control, perception, SLAM, Nav2 — then the merge ladder.